Lecture 7: Non-Linear Time Series Methods

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Beyond the SVMA

- We studied **linear SVMA models** because they are very attractive conceptually ...
  - Essentially every linearized structural model in the impulse-propagation paradigm can be written as a linear SVMA:
$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}$$
  - Knowing the Θ's = full characterization of the model: shock/policy causal effects, historical decompositions, business-cycle decompositions, ...
- ... but of course not particularly plausible. Two **Q**'s:
  1. True DGP of the world is **non-linear**. What do **linear strategies** give us in that case?
  2. Some economic questions are inherently **about non-linearities**: state & sign dependence, threshold effects, time variation [e.g., in policy], interaction effects, ... How to answer those?
     Example: maybe NKPC is non-linear—flat up to full employment, then vertical?

Non-linear estimation strategies

- **Impulse-propagation paradigm**: structural models are mappings

$$y_t = g(\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$$

  - Model is still a mapping from histories of shocks into macroeconomic outcomes, but now the function $g(\bullet)$ can be arbitrarily non-linear
  - Objective: characterize the function $g(\bullet)$ in as much detail as possible
  - Problem: different models imply very different types of f'n $g(\bullet)$. How should we proceed?

- Next few slides: quick summary of two recent advances

  1. Characterize estimand of simple **linear/non-linear LP strategies** under arbitrary DGP non-linearity [Main reference is Kolesár and Plagborg-Møller (2025).]
  2. Study **non-linear LPs** in special DGP: linear + time-varying policy
     [Main reference is Manuel and Wolf (2026).]

Outline

1. Potential Outcomes and LPs

2. Policy Rule Changes

3. Summary

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Bringing microeconometric insights to macro

Recent push: import insights from **cross-sectional treatment effect literature**

- How should we think about non-linearity/potential outcomes in a time series context?
  References are Rambachan & Shephard (2021) and Kolesár and Plagborg-Møller (2025).
  - Introduce general **potential outcomes framework** to a time series context
  - **Main takeaways**
    1. If observe outcomes and shocks (or IVs for the shock, plus a natural monotonicity condition): easily & robustly estimate weighted average treatment effects, for arbitrary non-linearity
    2. Statistical identification through heteroskedasticity and non-Gaussianity is extremely fragile – it *inherently* relies on linearity
       Also some concerns for identification relying on controls (= invertibility-based schemes). Sensitive to functional form of the included control set.
- I will elaborate in greater detail on 1.

Potential outcomes framework

- A **potential outcomes framework** for time series
  - Notation: scalar outcome y_t, scalar shock/policy shock $\varepsilon_{1,t}$, causal effect horizon of interest h, vector $u_{h,t}$ that collects all other variables affecting y_{t+h}, and orthogonal to $\varepsilon_{1,t}$
  - We can then define two key objects:
    1. *Potential outcome function*

       $$y_{t+h} = g_h(\varepsilon_{1,t}, u_{h,t})$$

       Gives the outcome as a function of the shock and everything else that affects the outcome.
    2. *Average structural function*

       $$G_h(\varepsilon) = \mathbb{E}\left[y_{t+h} \mid \varepsilon_{1,t} = \varepsilon\right] = \mathbb{E}\left[g_h(\varepsilon_{1,t}, u_{h,t}) \mid \varepsilon_{1,t} = \varepsilon\right]$$

       This function gives the counterfactual average value of y_{t+h} if the shock/policy was equal to ε, averaging out over the randomness caused by everything else independently affecting y_{t+h}.
- **Objective**: learn as much as possible about the average structural function $G_h(\bullet)$

Learning about potential outcomes

- We will assume for now that you observe the shock $\varepsilon_{1,t}$
  - The average structural function is then non-parametrically identified as

    $$G_h(\varepsilon) = \mathbb{E}\left[y_{t+h} \mid \varepsilon_{1,t} = \varepsilon\right]$$

    Could be estimated with enough (i.e., a huge amount of) data.
  - In practice: probably hopeless to fully characterize $G_h(\varepsilon)$, in particular if want to additionally condition on something else, e.g., state of the economy
  - We thus instead ask: can we give an interpretation to the estimand of *simple* estimation strategies, like simple LPs?
- Note: very similar conclusions if instead we only observe an IV for $\varepsilon_{1,t}$

  Need an additional monotonicity condition. See Proposition 3 in Kolesár and Plagborg-Møller (2025).

Linear methods in non-linear models

- Suppose we observe $\varepsilon_{1,t}$ and run the **linear local projection**

$$y_{t+h} = \theta_h^{\text{LP}} \varepsilon_{1,t} + \text{controls} + \text{error}$$

- Then what do we estimate for an arbitrary general non-linear DGP?

  A: You in general estimate a particular weighted average of causal effects for small shocks.

  - Again let $G_h(\varepsilon)$ denote the average counterfactual future outcome for y if the shock—e.g., a policy instrument—was set to some value ε: $G_h(\varepsilon) \equiv \mathbb{E}\left[y_{t+h} \mid \varepsilon_{1,t} = \varepsilon\right]$
  - Then, as we will show on the next slide:

$$\theta_h^{\text{LP}} = \frac{\text{Cov}(y_{t+h}, \varepsilon_{1,t})}{\text{Var}(\varepsilon_{1,t})} = \lim_{\delta \to 0} \int_{-\infty}^{\infty} w(\varepsilon) \frac{G_h(\varepsilon + \delta) - G_h(\varepsilon)}{\delta} d\varepsilon$$

    where $w(\varepsilon)$ is a weight function (non-negative and integrates to one)

  - Practical takeaway: plot $w(\varepsilon)$. Most weight on positive/negative/small/large shocks?

Linear methods in non-linear models

- Let $F(\bullet)$ denote the cdf of $\varepsilon_{1,t}$, assume mean zero. Begin with the numerator:

$$\begin{aligned}
\text{Cov}(y_{t+h}, \varepsilon_{1,t}) &= \mathbb{E}\left[\mathbb{E}\left(y_{t+h} \mid \varepsilon_{1,t}\right)\varepsilon_{1,t}\right] = \mathbb{E}\left[G_h(\varepsilon_{1,t})\varepsilon_{1,t}\right] \\
&= \int_{-\infty}^{\infty} G_h(\varepsilon)\varepsilon dF(\varepsilon) = \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\varepsilon} G_h'(a)da\right)\varepsilon dF(\varepsilon) \\
&= \int_{-\infty}^{\infty}\underbrace{\left(\int_{-\infty}^{\infty}\mathbf{1}_{a\leq\varepsilon}\varepsilon dF(\varepsilon)\right)}_{=\text{Cov}(1_{a\leq\varepsilon},\varepsilon)} G_h'(a)da
\end{aligned}$$

- Next consider the denominator:

$$\text{Var}(\varepsilon_{1,t}) = \text{Cov}(\varepsilon_{1,t}, \varepsilon_{1,t}) = \int_{-\infty}^{\infty}\text{Cov}(\mathbf{1}_{a\leq\varepsilon}, \varepsilon)da$$

- The overall weight function is thus

$$w(a) \equiv \frac{\text{Cov}(\mathbf{1}_{a\leq\varepsilon}, \varepsilon)}{\int_{-\infty}^{\infty}\text{Cov}(\mathbf{1}_{b\leq\varepsilon}, \varepsilon)db}$$

Note: this is the classical Yitzhaki (1996) weighted-average-of-slopes representation of OLS. If $\varepsilon_{1,t}$ is mean-0 and normal then it can be shown that $w(a) \propto f(a)$.

Weight functions in practice

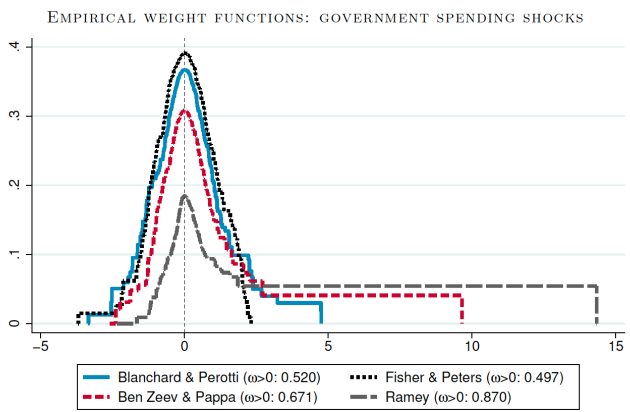


Figure 1: Estimated causal weight functions ω_X for government spending shocks obtained from the replication files for Ramey (2016), quarterly data. Horizontal axis in units of standard deviations. "$\omega > 0$:" total weight $\int_0^\infty \omega_X(x)\,dx$ on positive shocks. Papers referenced: Blanchard and Perotti (2002), Fisher and Peters (2010), Ben Zeev and Pappa (2017), Ramey (2011).

Simple non-linear methods in non-linear models

- Now suppose we observe $\varepsilon_{1,t}$ and run the simple ***non-linear* local projection**

$$y_{t+h} = \theta^{\text{LP}}_{0,h}(1 - s_{t-1})\varepsilon_{1,t} + \theta^{\text{LP}}_{1,h} s_{t-1}\varepsilon_{1,t} + \text{controls} + \text{error}$$

  - Interpretation: s_{t-1} here would typically be the state of the economy, e.g. recession versus expansion. E.g. see Ramey-Zubairy (2018).

- Then what do we estimate for an arbitrary general non-linear DGP?
  - Assume that $\varepsilon_{1,t}$ is also independent of s_{t-1}. Then by a similar argument to above:

$$\theta^{\text{LP}}_{s,h} = \lim_{\delta \to 0} \int_{-\infty}^{\infty} \tilde{w}(\varepsilon, s) \frac{\tilde{G}_h(\varepsilon + \delta, s) - \tilde{G}_h(\varepsilon, s)}{\delta} d\varepsilon$$

    where $\tilde{G}_h(\varepsilon, s) \equiv \mathbb{E}\left[y_{t+h} \mid \varepsilon_{1,t} = \varepsilon, s_{t-1} = s\right]$ and $\tilde{w}(\varepsilon, s)$ now additionally conditions on s

  - Takeaway: we now estimate a weighted average of state-dependent causal effects

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From rule changes to policy causal effects

- Remember Lecture Note 4: we can use **policy causal effects** to predict the effects of changes in **policy rules**. Obvious **Q**: couldn't we invert this logic?
  - Observe IRFs for a given shock $\varepsilon_{i,t}$ under two rules, denoted Θ_{ε_i} and $\bar{\Theta}_{\varepsilon_i}$. Compute

$$\Theta_{\varepsilon_i} - \bar{\Theta}_{\varepsilon_i}$$

  - But it follows immediately that this recovers **policy causal effects**! Specifically:

$$\Theta_{x,\varepsilon_i} - \bar{\Theta}_{x,\varepsilon_i} = \underbrace{\Theta_{x,z}}_{\text{instrument causal effects}} \times \underbrace{\left(\Theta_{z,\varepsilon_i} - \bar{\Theta}_{z,\varepsilon_i}\right)}_{\text{treatment}}$$

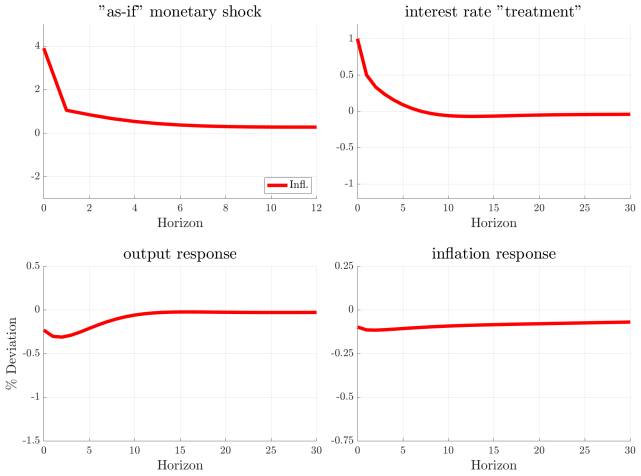
    Note: $\Theta_{x,z} = \Theta_{x,\nu}\Theta_{z,\nu}^{-1}$ is exactly the instrument-space causal effect matrix of Lecture Note 4.

    1. Note: comparing each shock $\varepsilon_{i,t}$ across two regimes identifies a different policy treatment
    2. If shocks and regimes are persistent, then this treatment is likely to be persistent. Quite different from standard policy shocks that we saw in Lecture Note 5.

- Next few slides: sketch of main idea & application. Main reference: Manuel and Wolf (2026).

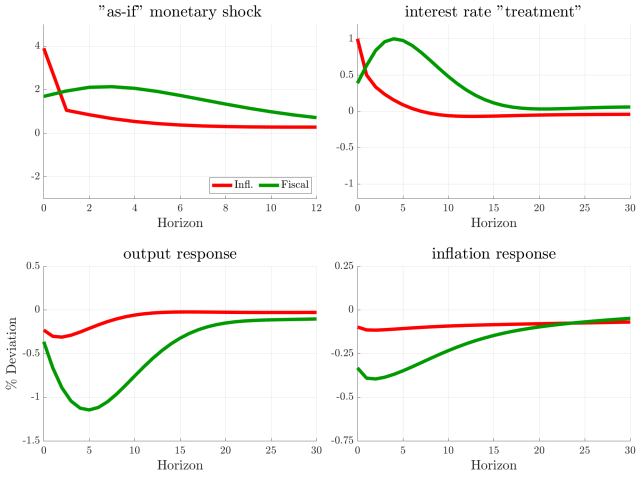
Heterogeneous treatment effects

Illustration: shock transmission Δ across monetary policy rules in Smets-Wouters (2007)



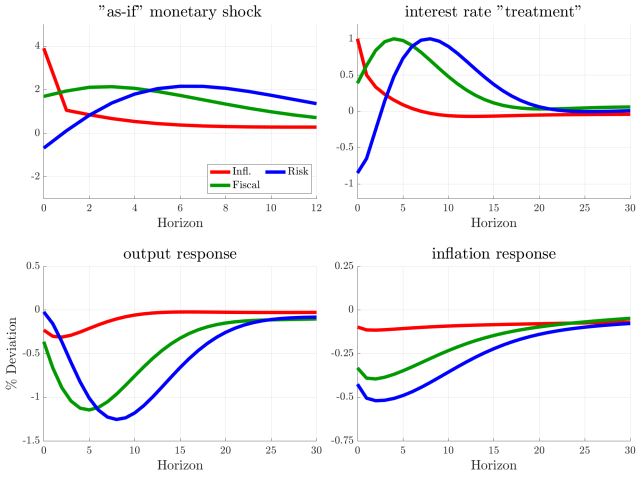
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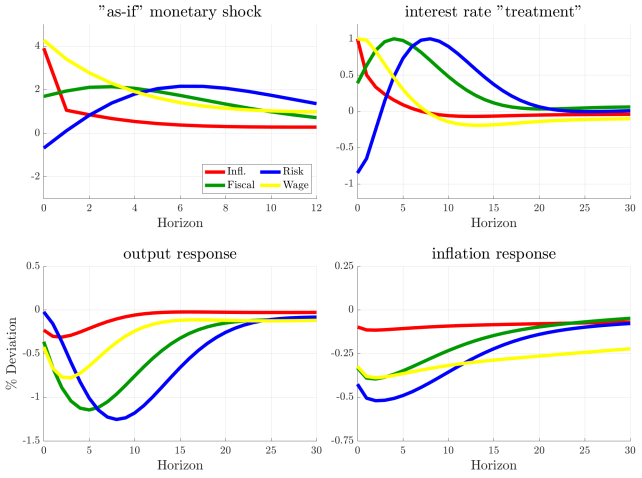
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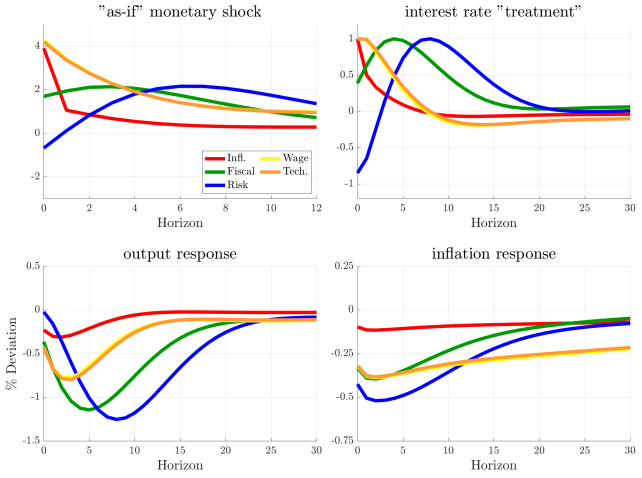
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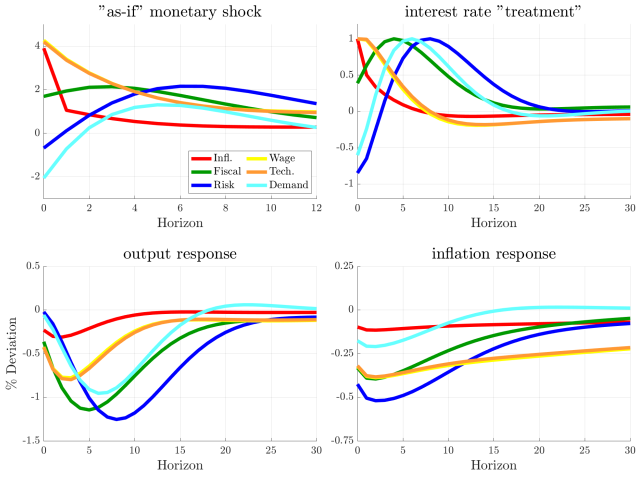
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Empirical specification

- Run interacted local projections of the form

$$y_{t+h} = \text{constant} + \beta_{y,h}\varepsilon_{i,t} + \gamma_{y,h}\zeta_t + \delta_{y,h}\left(\varepsilon_{i,t} \cdot \zeta_t\right) + \text{error}.$$

  - Outcomes y: policy instrument z and macro outcome of interest x
  - Regressors:
    a) $\varepsilon_{i,t}$: non-policy shock [e.g., oil]
    b) ζ_t: correlate of systematic design of instrument z, and not of other determinants of propagation of ε [e.g., changes in composition of FOMC due to mechanical rotation]

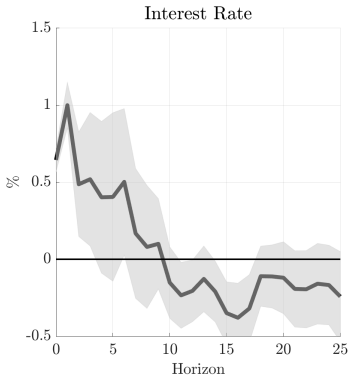
- Main result of Manuel and Wolf (2026): conditions on DGP & regressors $\varepsilon_{i,t}$, ζ_t so that

$$\boldsymbol{\delta}_x = \Theta_{x,z} \cdot \boldsymbol{\delta}_z.$$

Running the LP for z gives the treatment, for x we get the associated causal effects.

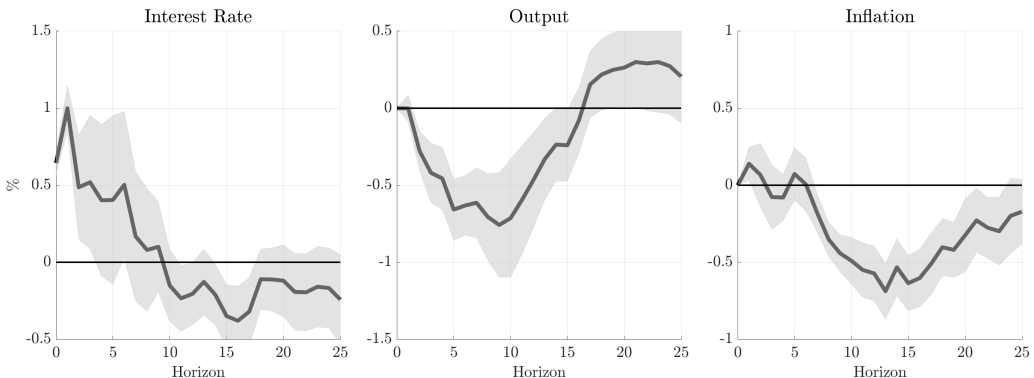
Application

recap: the standard **monetary shock** of Romer-Romer (2004)



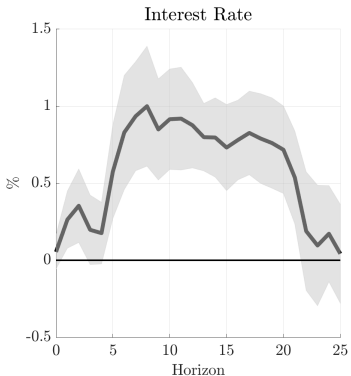
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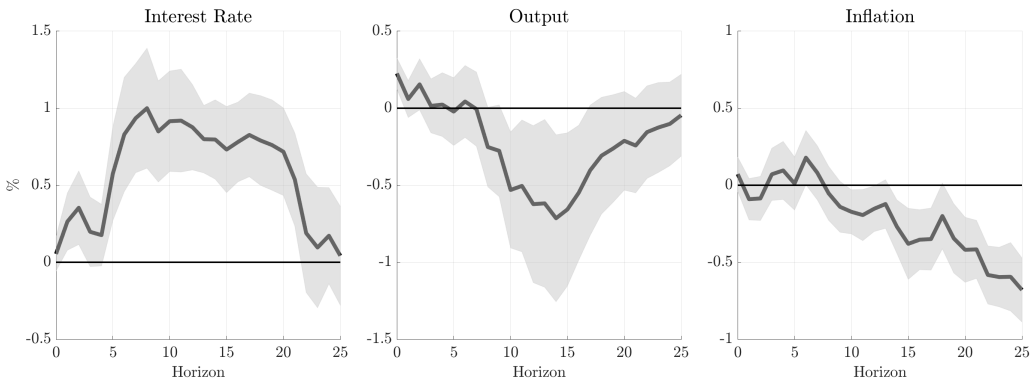
Application

FOMC composition [Hack-Istrefi-Meier (2024)] × **Hamilton oil shock**



Application

FOMC composition [Hack-Istrefi-Meier (2024)] × **Hamilton oil shock**



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Summary & takeaways

- This lecture: **non-linear** time series estimation methods
  1. Linear LPs with identified shocks/IVs admit **tight causal interpretations** even in non-linear DGPs; the same is true for simple non-linear LPs
  2. Can use time variation in policy as a source of **identifying information**
- Some interesting **open questions**:
  - Do non-linearities matter in the aggregate? Which are most important?
    E.g., see Auclert-Rigato-Rognlie-Straub (2024) for a discussion of NKPC non-linearities in structural menu cost models. How big are non-linearities in the data?
  - What kind of policy treatments does time variation in policy identify? Can we use it to increase our understanding of policy dynamic causal effects?
- Next up: three (short) lecture notes on **applications**